THE NOVA RESEARCH CURRICULUM

Physics

Grade 7 · Term 1

Student edition

Outcome: PH-7.1 speed, distance–time graphs, contact and non-contact forces, action and reaction.

Lesson notes

- **Speed (Weeks 1–4):** speed = distance ÷ time; units m/s and km/h. Distance–time graphs: a straight sloping line = steady speed; flat line = stopped; steeper = faster. Practical: time a walk over 20 m.

  ▶ **Try it on NovaSchool:** Moving car with a live distance–time graph — open **Simulations** on your dashboard.

- **Forces (Weeks 5–8):** a force is a push or pull, measured in newtons (N). Contact: friction, air resistance, tension, normal reaction. Non-contact: gravity, magnetic, electrostatic. Force diagrams with arrows.
- **Action and reaction (Weeks 9–12):** every action has an equal and opposite reaction (Newton's third law): a balloon rocket, swimming, a rocket launch.

  ▶ **Try it on NovaSchool:** Balloon rocket: action and reaction — open **Simulations** on your dashboard.

Workbook pages

Page	Task
1–2	Speed calculations; draw and read distance–time graphs.
3	Sort 12 forces into contact and non-contact; force diagrams.
4	Balloon-rocket practical report.

Question bank

#	Outcome	Question	Your answer
1	PH-7.1	A car goes 120 km in 2 hours. Its speed?	240 km/h / 60 km/h / 122 km/h / 118 km/h
2	PH-7.1	A flat line on a distance–time graph means the object is …	speeding up / not moving / moving steadily / slowing down
3	PH-7.1	Force is measured in …	kilograms / metres / newtons / joules
4	PH-7.1	Which is a non-contact force?	friction / tension / gravity / air resistance
5	PH-7.1	Friction always acts against the direction of motion. True or false.	True / False

#	Outcome	Question	Your answer
6	PH-7.1	When you push a wall, the wall pushes back on you. True or false.	True / False
7	PH-7.1	Explain how a balloon rocket moves.	_____
8	PH-7.1	A runner covers 100 m in 20 s. Speed?	5 m/s / 2 m/s / 120 m/s / 2,000 m/s